

Supplementary Information for *Activation Patching in Structured CNNs: Intervention Size, Architecture, and Additivity*

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S1 Architecture and training specification

For each task and renderer block, the data generator draws 128 training, 64 validation, and 64 test nuisance contexts from independent deterministic streams. Each context is instantiated in all four class states, giving 512/256/256 images in the three splits. Models are trained for 200 epochs with Adam, learning rate 0.003 and batch size 128. Retention keeps the first-layer penalty at $\lambda = 1$ throughout training. Release uses the same penalty through epoch 10, decreases it linearly to zero at epoch 80, and leaves it at zero thereafter. Within the fresh-sample architecture study, paired retention/release runs use the same model initialization and paired minibatch-order streams for a given renderer block, task, architecture, and initialization replicate.

All architectures share a bias-free $1 \rightarrow 16$, 9×9 first convolution, same padding, ReLU, and the same first-layer intervention tensor. GMP and GAP denote global max and global average pooling. Plain depth-2 models add one downstream 3×3 convolution; plain depth-4 models add three. Width-64 variants expand the first downstream convolution from 16 to 64 channels and remain width 64 thereafter. Residual depth-4 models remain width 16 and add the third downstream convolution to the original first-layer activation before the final ReLU. BatchNorm, when present, occurs only downstream of the intervention tensor, after each downstream convolution. A learned linear classifier maps the pooled representation to four classes.

The exact architecture constructor, training loop, frozen design, and parameter counts are included in the accompanying reproducibility artifact under `studies/cnn_architecture_robustness/` and `analysis/architecture_robustness.001/training_design.json`.

Table S1 Complete 16-model architecture grid. Depth counts the shared first convolution.

Architecture	Depth	Width	Pool	Residual	BatchNorm	Parameters
tiny_gmp	1	16	GMP	no	no	1,364
tiny_gap	1	16	GAP	no	no	1,364
plain2_w16_gmp	2	16	GMP	no	no	3,684
plain2_w16_gap	2	16	GAP	no	no	3,684
plain4_w16_gmp	4	16	GMP	no	no	8,324
plain4_w16_gap	4	16	GAP	no	no	8,324
plain2_w64_gmp	2	64	GMP	no	no	10,836
plain2_w64_gap	2	64	GAP	no	no	10,836
plain4_w64_gmp	4	64	GMP	no	no	84,692
plain4_w64_gap	4	64	GAP	no	no	84,692
res4_w16_gmp	4	16	GMP	yes	no	8,324
res4_w16_gap	4	16	GAP	yes	no	8,324
bn2_w16_gmp	2	16	GMP	no	yes	3,700
bn2_w16_gap	2	16	GAP	no	yes	3,700
bn4_w16_gmp	4	16	GMP	no	yes	8,372
bn4_w16_gap	4	16	GAP	no	yes	8,372

S2 Intervention-size and metric robustness

For a patching metric M , the main intervention-size contrast is

$$B^M = \frac{\Delta^M(4) + \Delta^M(8)}{2} - \frac{\Delta^M(1) + \Delta^M(2)}{2}, \quad (\text{S1})$$

where $\Delta^M(k)$ is release minus retention at intervention size k . The initial pre-specified test uses 20 held-out renderer blocks and the primary **two_concepts**/TinyCNN setting.

Table S2 Primary intervention-size contrast under the historical and two alternative evaluation metrics. Intervals are paired 95% intervals over the 20 held-out renderer blocks.

Metric	B	95% CI
Probability fidelity	+0.33098	[0.27368, 0.38827]
Centered-logit fidelity	+0.17888	[0.12727, 0.23049]
Probability-error reduction	+0.89441	[0.78630, 1.00252]

The alternative metrics reuse the frozen checkpoints and validation-derived channel rankings. Their metric definitions and analysis plan were fixed before evaluating the alternative-metric outcomes. Full pointwise curves and block-level values are included in **analysis/metric_sensitivity_001/** in the accompanying artifact.

S3 Architecture robustness

The fresh-sample architecture study uses 100 renderer blocks, four paired initialization replicates per block, the 16 architectures in Table S1, two tasks, and the two training regimes. Initialization replicates are averaged within renderer block before inference.

Table S3 Repeated-measures architecture omnibus tests on the primary `two_concepts` task.

Metric	F	df	partial η^2	Holm p
Centered-logit fidelity	77.09	15, 1485	0.438	2.40×10^{-173}
Probability-error reduction	656.70	15, 1485	0.869	$< 10^{-300}$

The architecture-specific estimates plotted in the main paper are stored in `analysis/architecture_robustness_001/architecture_B_summary.csv`. The corresponding training design, factor contrasts, input-effect diagnostics, and omnibus results are included alongside that file. Architecture comparisons are not accuracy-matched and are interpreted as properties of the trained systems in this controlled grid.

After the pre-specified architecture results were known, an explicitly post-hoc robustness analysis checked the omnibus conclusion under sphericity correction, within-block permutation, rank-based Friedman tests, and random channel orders. These checks support the robustness statement in the main text without changing the pre-specified status of the original architecture test.

Table S4 Post-hoc robustness checks for architecture heterogeneity on `two_concepts`. Permutation values are Holm adjusted across the two metrics.

Metric	GG ϵ	GG p	Permutation Holm p	Friedman p
Centered-logit fidelity	0.1861	1.56×10^{-34}	2.0×10^{-5}	3.52×10^{-146}
Probability-error reduction	0.5973	$< 10^{-300}$	2.0×10^{-5}	7.79×10^{-200}

S4 Pixel-permuted spatial control

The spatial-control experiment compares the original structured filter bank with a bank obtained by applying one common permutation to the 81 spatial coordinates of every filter. This transformation preserves row norms, the complete Gram matrix, numerical rank, singular spectrum, and row coefficient multisets, while destroying the original two-dimensional arrangement. The primary analysis uses 100 new renderer blocks and four architectures.

Table S5 Architecture-averaged structured-minus-pixel-permuted contrast for validation-selected channels on **two_concepts**. Difference intervals are paired Student-*t* 95% intervals from the primary spatial-control analysis; equivalence uses the pre-specified 90% TOST interval and symmetric margin.

Metric	Mean difference	95% CI	90% CI	Margin
Centered-logit fidelity	+0.05743	[0.05116, 0.06369]	[0.05219, 0.06266]	± 0.01704
Probability-error reduction	+0.04563	[0.03728, 0.05397]	[0.03864, 0.05261]	± 0.06594

The centered-logit interval lies outside its equivalence region. The probability-error 90% interval lies inside its wider study-specific equivalence margin even though its ordinary difference interval excludes zero.

Table S6 Structured-minus-pixel-permuted *B* shown by architecture on **two_concepts**.

Architecture	Centered-logit difference	Probability-error difference
tiny_gmp	+0.10624	+0.19953
tiny_gap	+0.07690	+0.04308
plain2_w16_gmp	-0.04453	-0.08067
plain2_w16_gap	+0.09109	+0.02057

The sign reversal in **plain2_w16_gmp** is why the main text limits the spatial conclusion to an architecture- and metric-dependent effect rather than a universal positive contribution of the original bank.

S5 Singleton-additive decomposition

The decomposition re-evaluates saved checkpoints one first-layer channel at a time and reconstructs a joint centered-logit intervention from the sum of singleton centered-logit changes. It evaluates all $k = 0, \dots, 16$ while retaining $k = 1, 2, 4, 8$ for the definition of *B*. The protocol, implementation manifest, numerical unit tests, and analysis code are included under **studies/cnn_patching_additivity/**.

Table S7 Primary examples from the singleton-additive analysis on **two_concepts**. The BN2-GAP paired interval is computed over the same 100 renderer blocks.

Architecture / metric	Observed <i>B</i>	Additive <i>B</i>	Observed – additive
tiny_gmp , centered logits	+0.14558	+0.14558	numerical precision
tiny_gmp , probability error	+0.83151	+0.83151	numerical precision
bn2_w16_gap , centered logits	-0.13817	+0.04177	-0.17994

For `bn2.w16_gap`, the paired centered-logit difference is -0.17994 with 95% Student- t interval $[-0.20082, -0.15906]$. In the analytic `tiny_gmp` control, the aggregate centered residual-energy ratio remains at floating-point scale: 8.41×10^{-14} for release and 1.45×10^{-13} for retention at $k = 16$. For `bn2.w16_gap` at $k = 8$, the corresponding ratios are 0.190 and 0.339.

Table S8 Observed and additive reconstruction of the architecture-averaged structured-minus-pixel-permuted B on `two_concepts`. The final column is the paired observed-minus-additive comparison over 100 renderer blocks.

Metric	Observed	Additive	Paired difference [95% CI]
Centered-logit fidelity	+0.05743	+0.05710	+0.00033 $[-0.00074, +0.00140]$
Probability-error reduction	+0.04563	+0.04105	+0.00458 $[+0.00202, +0.00714]$

These comparisons support the narrower claim made in the article: the singleton reconstruction reproduces the studied architecture-averaged spatial contrast closely, but architecture-specific reconstruction gaps remain.

S6 Reproducibility map

Table S9 identifies the exact presentation script and archived analysis tables used for each main figure. The accompanying reproducibility artifact preserves these paths relative to the repository root.

Table S9 Main-figure provenance.

Figure	Presentation script	Principal archived inputs
1	<code>paper/build_overview_figures.py</code>	<code>studies/cnn_release_experiment/core.py</code>
2	<code>paper/build_main_results_figures.py</code>	<code>analysis/metric_sensitivity_001/stage_d_treatment_contrasts.csv</code> ; <code>analysis/architecture_robustness_001/architecture_B_summary.csv</code>
3	<code>paper/build_anchor_specificity_figures.py</code>	<code>analysis/anchor_specificity_001/secondary_anchor_contrasts.csv</code> ; <code>primary_spatial_specificity.csv</code> ; <code>random_channel_spatial_specificity.csv</code>
4	<code>paper/build_additivity_curve_figures.py</code>	<code>analysis/patching_additivity_001/architecture_analysis/curve_summary.csv</code>
5	<code>paper/build_additivity_figure.py</code>	<code>analysis/patching_additivity_001/architecture_analysis/B_summary.csv</code> ; <code>analysis/patching_additivity_001/anchor_analysis/structured_vs_pixel_permut</code>

The paired BN2-GAP interval in Section S5 is reproduced from `architecture_analysis/B_per_block.csv`. The paired spatial observed-minus-additive intervals are reproduced from `anchor_analysis/B_per_block.csv`. The frozen protocols corresponding to the intervention-size, metric, architecture, spatial-control, and additivity analyses are included in the artifact.

S7 Integrity checks and artifact scope

Full-patch and no-op identity checks are recorded in the archived evaluator summaries. For the analytic Tiny controls, observed and singleton-additive centered-logit curves agree to floating-point precision. An independent checkpoint reimplementation audit of the earlier causal evaluator is also included as `analysis/checkpoint_audit/AUDIT_REPORT.md`.

The submission artifact intentionally excludes raw training checkpoints and tens of thousands of per-model evaluator JSON files. Those files are retained separately because of their size. The included block-level and aggregate CSV files are sufficient to regenerate the main figures and the reported block-level inferential summaries without retraining the models. A file-by-file map and package-building instructions are provided in `paper/SUPPLEMENT_REPRODUCIBILITY.md`.